Lepton Masses from Rope Topology

A Koide-Phase Model: Numerical Success, and Its Pinned Failure Mode

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Naming note (5 September 2026). This programme is now the Mesh Programme and the theory it develops the mesh framework; 'the rope framework' in earlier revisions reads 'the mesh framework'. 'Rope' remains the name of the two-strand wound object, and the Rope Hypothesis credited to Bill Gaede keeps its name. Identifiers (repository, rope_solver, file names, claim IDs) are unchanged.

Abstract

The mesh framework suggests a topological origin for the charged-lepton mass ratios through a Koide-type phase relation with coefficient (3+Φ) = 4.618034, where Φ is the golden ratio. Used with the measured Weinberg angle, the relation reproduces the lepton mass ratios to within about one percent: μ/e = 205.8 (measured 206.8) and τ/μ = 16.82 (measured 16.82). We report this success and, with equal prominence, its failure mode: with the rope model's own predicted Weinberg angle the ratios are badly wrong. The result is a numerically striking CONJECTURE conditional on a measured input, not a derivation. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants; the lepton-ratio caveat below is pinned in the test suite.

> **Scope of this paper**
>
> CLAIM: a Koide-phase relation with coefficient (3+Φ)=4.618 reproduces the charged-lepton mass ratios to ~1% WHEN the measured $\sin^2\theta_W$ is used.
>
> NOT CLAIMED: a first-principles derivation. With the model's own $\sin^2\theta_W$ the ratios fail; the success is conditional on a measured input. Status: DEMOTED 5 September 2026 (KOIDE-WEINBERG, PM-001) to a KEPT COINCIDENCE: the sensitivity to the input angle is 71. See the Disposition section.

1. The Koide Phase and Its Coefficient

The empirical Koide relation among charged-lepton masses is reproduced in a phase parametrisation whose coefficient the mesh framework computes as (3+Φ) = 4.618034, with Φ the golden ratio 1.618034. This coefficient is the model's structural contribution.

2. Numerical Result

Evaluated with the measured Weinberg angle $\sin^2\theta_W = 0.23122$, the relation yields:

Ratio	Rope (measured θ_W)	Experiment	Agreement
μ/e	205.8	206.8	~0.5%
τ/μ	16.82	16.82	~0.04%
τ/e	3462.7	3477.2	~0.4%

The pinned honest caveat. The lepton mass ratios come out close to experiment ONLY when the measured Weinberg angle sin²θ_W = 0.23122 is used as input. With the rope model's OWN predicted value sin²θ_W = 1/(3√2) = 0.2357 — itself about 1.9% from the measured value — the ratios fail badly (μ/e jumps from ~206 to ~1605). The mass-ratio success is therefore CONDITIONAL on an externally measured input the model does not itself reproduce accurately. This is a conjecture with a known failure mode, not a derivation, and it is recorded as such in the test suite.

Disposition (5 September 2026): demoted to a kept coincidence

The sensitivity, computed before any demotion was discussed. Under this paper's exact construction (√m_k proportional to 1 + √2 cos(φ + 2πk/3) with φ = (3+Φ)θ_W, pinned in rope_solver.particles), d ln(m_μ/m_e) / d ln(sin²θ_W) = 71 at the measured angle, by exact algebra with the angle carried as a symbol and confirmed by finite difference. A 1.94% change in the input moves the muon-to-electron ratio by a factor 7.8. Commission KOIDE-WEINBERG (ITEM 6d) had fixed a threshold of 10 before computing: a relation whose output moves seventyfold per unit relative input is a fit to the input, not a prediction. That is exactly what the pinned failure mode (PM-002) had been saying.

The dependency path. The chain is Koide's empirical parametrisation (external to the framework) plus the identification φ = C θ_W plus C = $3d_0 + d_1/_2$ at level k = 3. The $d_1/_2$ = Φ part is rigorous; the '3 + 1 T-parity mode count' behind C is conjectural by the construction's own docstring; the identification of the Koide phase with the Weinberg angle is a choice. Three choice points and the sensitivity of 71: the registered verdict is PM-DEMOTE. PM-001 leaves Conjecture grade and is kept as a coincidence; PM-002 stands as the pinned failure. Full record: analysis/KOIDE_WEINBERG_results.md.

What stands, and what the corpus now says. The (3+Φ) coefficient stands as the golden-ratio structure it was derived from; the numerical landing with the measured angle stands as a number. The Koide relation's status in the corpus is 'an observed relation among measured masses, not a rope result'. PM-003 (the lepton problem is a three-level excitation spectrum of one knot) is untouched: it says a relation among levels is the right kind of tool, not that this one is it. The Weinberg-angle estimate this relation depended on (EW-001) was demoted in the same commission; the coupled inconsistency between the two entries is closed.

Status

Koide coefficient (3+Φ) = 4.618 — Derived (from golden-ratio structure).

Lepton ratios to ~1% with MEASURED θ_W: DEMOTED to a kept coincidence (PM-001, 5 September 2026); sensitivity d ln(m_μ/m_e)/d ln(sin²θ_W) = 71.

Ratios with the model's OWN θ_W — Failed (μ/e ~1605); pinned in tests.

Registered Claims in This Paper
Each statement below is traceable to the machine-readable registry (claims.yaml), the corpus's single source of truth. Status labels: Derived (follows from stated mechanics), Modeled (reproduces data with declared inputs), EFT-constrained (bounded by effective-theory limits), Conjecture (proposed, not derived), Open (registered unsolved problem), Failed (falsified, kept as a finding). Where a claim is code-backed, the named benchmark reruns it.

- **PM-001 [Failed: demoted 5 Sep 2026, kept coincidence]:** Lepton mass ratios via Koide phase

- **PM-002 [Failed]:** Lepton ratios with the model's own Weinberg angle

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.